Database Systems Project Proposal

Project Title

Course Registration & Management System

Group Members & Course Information

Course: Database Systems

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Introduction / Background

The **Course Registration & Management System** is a database-driven application designed to efficiently manage and automate the course enrollment process in academic institutions. Traditional registration systems often rely on manual verification or loosely structured data handling, which leads to inconsistencies and errors.

This project focuses on designing a **robust relational database system** where academic policies and constraints are enforced directly through schema design, relationships, and integrity rules. The system ensures that all operations—such as course registration, prerequisite validation, and billing—are handled accurately and consistently using database-level logic.

Problem Statement

Existing course registration processes often suffer from:

- Lack of enforcement of prerequisite constraints
- Students exceeding allowed credit hour limits
- Over-enrollment in course sections
- Redundant or inconsistent data storage
- Inefficient fee calculation and tracking

These issues arise due to weak database design and insufficient constraint enforcement. The goal of this project is to eliminate these problems by building a **well-structured, normalized database system** with strict integrity constraints.

Objectives of the Project

- Design a **fully normalized relational database schema (up to 3NF)**
- Implement **entity relationships with referential integrity constraints**
- Enforce **academic rules using database constraints and logic**
- Ensure **data consistency and eliminate redundancy**
- Support efficient querying for registration, academic tracking, and billing
- Simulate real-world database operations such as transactions and updates

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Scope of the Project

The project primarily focuses on the **database design and implementation** aspects of the system. The scope includes:

- Schema design and normalization
- Implementation of constraints (primary keys, foreign keys, checks)
- Query design for registration and validation
- Transaction handling for course enrollment and billing updates

A basic frontend may be included for interaction, but the core emphasis remains on the **database layer and its functionality**.

System Architecture / Design

The system revolves around a **central relational database**, with the following actors:

- **Student:** Performs course registration and views records
- **Registrar (Admin):** Manages courses, sections, and users
- **Faculty:** Accesses assigned course data
- **Billing Office:** Handles financial records

Core Functionalities (Database Perspective):

Academic Constraints:

- Prerequisite enforcement using enrollment history
- Credit hour limit (maximum 18 credit hours per semester)
- Fail-repeat rule enforced through triggers
- Automatic lab-course linkage via relational mapping

Faculty & Section Constraints:

- Faculty assigned to one subject
- Maximum 3 sections per faculty
- Section capacity limited to 50 students

Financial Constraints:

- Fee calculation stored and computed via queries
- Payment status tracking using transactional updates

Tools and Technologies

- **Database:** PostgreSQL (Supabase)
- **Query Language:** SQL
- **Optional Frontend:** React.js (for interface)
- **Version Control:** Git & GitHub

Expected Results and Conclusion

The outcome will be a **fully functional relational database system** capable of handling course registration with strict enforcement of academic and financial rules.

By focusing on **database-level constraint management**, the system will ensure data accuracy, eliminate redundancy, and prevent invalid operations. This project highlights the importance of **proper database design in solving real-world problems** and demonstrates the practical application of core database concepts such as normalization, transactions, and integrity constraints.